Prof. Patti Hamerski

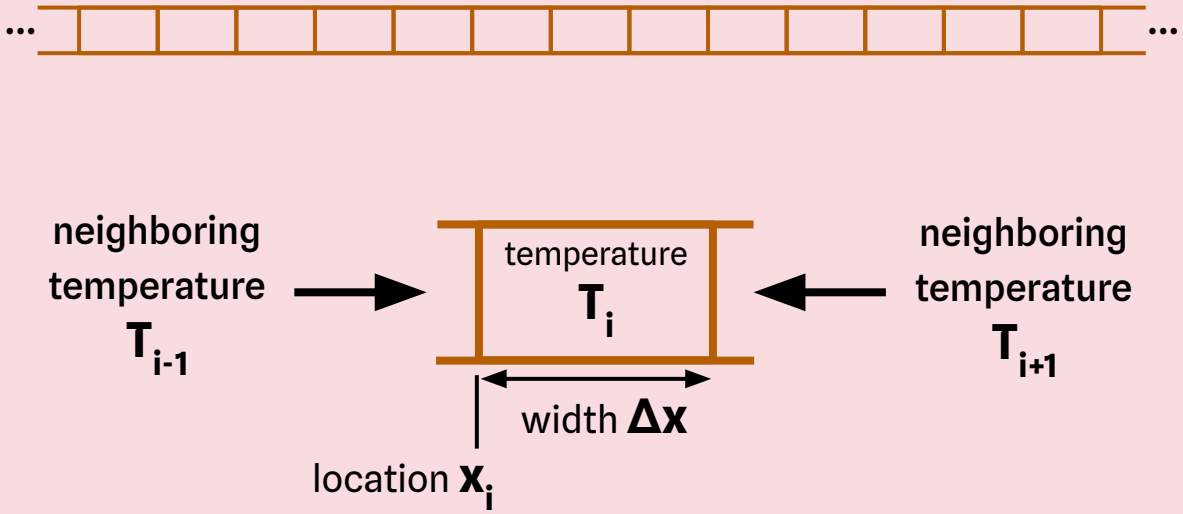
Day 2

Introduction to Finite Difference Methods

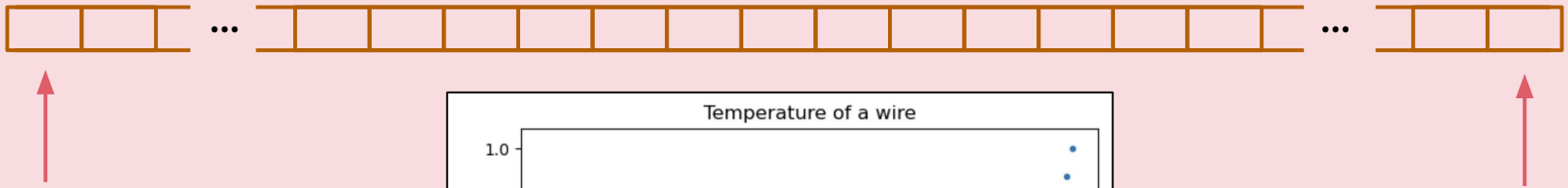
Spring 2025

PH 367: Computational Physics Lab III

Splitting a wire into chunks

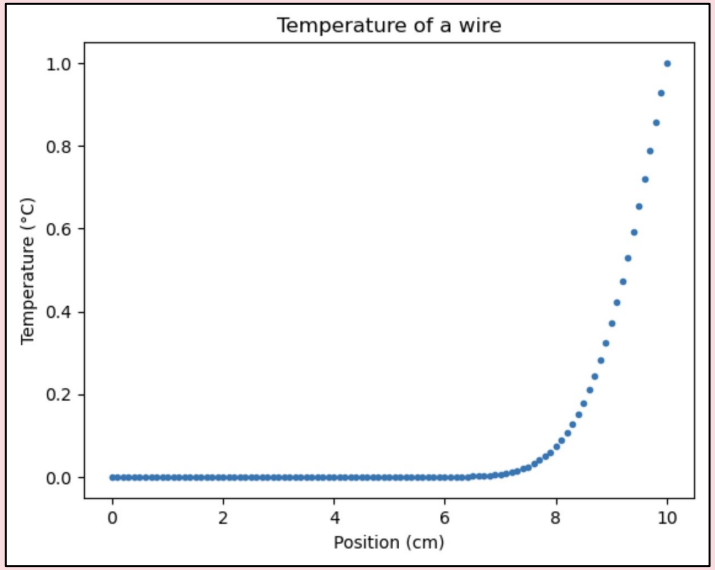


Splitting a wire into chunks



Held at 0 degrees

Held at 1 degree



Code for updating temperature

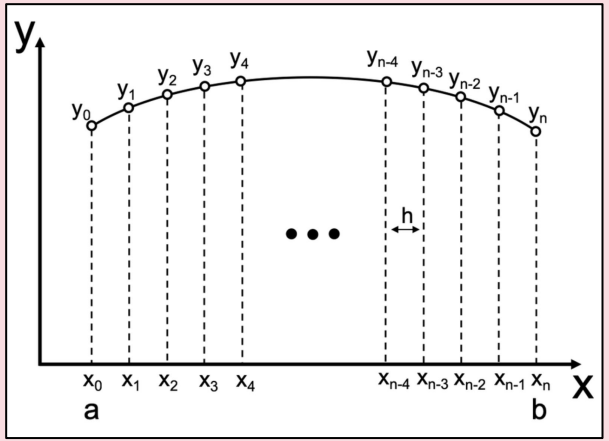
```
while t < tf:
    rate = np.zeros(N)
    for i in range(1, N-1):
        rate[i] = alpha * (T[i-1] - 2 * T[i] + T[i+1]) / dx**2
    dT = rate * dt
    T += dT
    t += dt
```

$$\frac{\mathrm{d}T_i}{\mathrm{d}t} = \alpha \frac{(T_{i-1} - 2T_i + T_{i+1})}{\mathrm{d}x^2}$$

Finite Difference Methods

One dependent variable

Two independent variables (usually x and t)



$$\frac{\partial y_i}{\partial x} = \frac{y_{i+1} - y_{i-1}}{2h}$$

$$\frac{\partial^2 y_i}{\partial x^2} = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2}$$

Finite Difference Methods

FDM formula for second derivative

$$\frac{\partial^2 y_i}{\partial x^2} = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2}$$

Heat equation

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\frac{\mathrm{d}T_i}{\mathrm{d}t} = \alpha \frac{(T_{i-1} - 2T_i + T_{i+1})}{\mathrm{d}x^2}$$